

BI Internship Assignment

Project: UTel Subscriber Intelligence & Usage Analytics Platform

Background

UTel would like to establish a Business Intelligence platform capable of transforming operational telecom data into management information.

The BI team has received operational data extracts from various internal systems. The data includes subscriber information, voice usage, SMS usage, airtime purchases, internet bundle purchases and fibre subscriptions.

Before management reports can be produced, the data must be designed, acquired, transformed, cleaned and loaded into a reporting database.

You are required to design and implement a complete BI pipeline from raw data to management reporting.

Provided Datasets

Subscribers.csv

Calls.csv

SMS.csv

AirtimeSales.csv

InternetBundleSales.csv

FibreSubscriptions.csv

Part A: Database Design

Study all datasets and design a database capable of storing and relating the data.

Tasks:

1. Identify entities.
2. Identify attributes.
3. Identify relationships.
4. Design an Entity Relationship Diagram (ERD).
5. Define primary keys and foreign keys.

Deliverables:

- ERD Diagram

Part B: Data Transformation & Data Quality Checks

Using Python perform a data quality check.

Tasks:

1. Remove missing values.
2. Remove duplicate records.
3. Standardize date formats.
4. Validate MSISDN values.
6. Remove invalid records.

Deliverables:

- Python Transformation Script

Part D: Data Loading

Load cleaned datasets into the designed MySQL database.

Tasks:

1. Create database tables.
2. Load transformed data into tables.

Deliverables:

- MySQL Database
- SQL Scripts
- Python Loading Script

Part E: Business Intelligence Analysis

Using SQL, Answer the following business questions.

Subscriber Growth

- How many subscribers exist?
- How many subscribers were acquired each month?
- What is the subscriber growth trend?

Subscriber Activity

- How many subscribers made calls?
- How many subscribers sent SMS?

- How many subscribers purchased airtime?
- How many subscribers purchased Internet?

SMS Usage

- Who are the Top 10 SMS users?

Voice Usage

- What is the average call duration?
- Who are the Top 10 callers?

Revenue Generation

- How much airtime revenue was generated?
- How much fibre revenue was generated?
- How much voice revenue was generated?
- What is the total revenue?

Part F: Customer Segmentation (10 Marks)

Using total customer revenue:

Total Revenue =
Airtime Revenue +
Bundle Revenue +
Voice Revenue +
Fibre Revenue

Create the following customer segments:

- Bronze
- Silver
- Gold
- Platinum

Produce a report showing:

- MSISDN
- Total Revenue
- Segment

Part G: SSRS Reporting

Develop SSRS reports and dashboards showing:

Subscriber Growth KPIs
Customer Segmentation KPIs

Deliverables:

- SSRS Dashboard